Personal Statement

Zhenghao Chen · PhD Application —
Graph Intelligence & Urban Resilience

My path into research began with a lifelong curiosity about geography. As a high
— first clarify the center of a problem, then think across its dimensions
—
and it eventually led me to the conviction that now defines my research: the stru

I entered the Jianxing Honors College at Zhejiang University of Technology, chos
–
diffusion equations, and studied deep-learning traffic-flow imputation with RNN
– cloud collaborative aquaculture monitoring system —
the first to apply SlowFast to fish-behavior detection (97.06% accuracy)
—
which was deployed in intensive American-shad farming and later accepted at II
“ Deep Learning for Time Series Forecasting: A Survey ”
(IJMLC 2025), now cited more than 420 times on Google Scholar, which taught

This intuition matured into a systematic study of graph neural networks across t
—
denoising noisy semantic graphs through behavior-conditioned diffusion (IGDM

Beyond model-centric questions, I extend this network perspective to problems w
’ s Infrastructure, AI and Urban Nexus Lab, I quantified how a city ’
s heat accumulation is shaped by the structure of its street network. This grew in
’
s heat-ventilation potential. Linking UTRI to socioeconomic data in Washington,
—
quantitative evidence that heat risk is an environmental-justice issue embedded

These experiences converge on the question I want to pursue in my PhD: how ca
— diffusion, reaction – diffusion —

with relational structure for spatiotemporal climate prediction; and (3) translate

I am drawn to the University of Auckland because this question sits at the interse

ā Ara Whet ū

Centre for Climate, Biodiversity and Society pursues precisely the evidence-base

’

s research on urban and infrastructure network resilience and cascading failure

— such as Dr. Conrad Zorn ’

s work on regional network resilience and post-disaster decision-making, and the

—

which align directly with my finding that street-network topology embeds enviro

’

s exposure to climate hazards, and its leadership in resilience science, make it the

My goal is to work on the front line of climate adaptation and disaster response,

—

I value persistence above all. I hope to carry this thread into climate-change rese